Peng Li

CFD | Fluid–Structure Interaction | Dynamic Analysis

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Professional Summary

Ph.D. researcher at KTH Royal Institute of Technology with expertise in computational fluid dynamics (CFD), fluid–structure interaction (FSI), and structural dynamics. Develop coupled numerical models to predict fluid loading, deformation, vibration, and transient system response. Combine high-fidelity simulation with analytical modelling, parameter studies, numerical verification, and comparison with measurements. Industrial experience includes Python-based model development and coupled OpenFOAM simulations supporting airbag-inflator design analysis. Seeking to apply these strengths to vessel dynamics and wave-energy engineering at CorPower Ocean. Ph.D. defence scheduled for 4 December 2026.

Technical Skills

Fluid dynamics and FSI: Compressible LES; transient CFD; finite-volume and finite-element methods; strong two-way fluid–structure coupling; fluid loading; overset and deforming meshes; flow-induced vibration; shock-wave propagation.

Structural and dynamic analysis: Hyperelastic material modelling; large deformation; structural eigenmodes and resonance; force–velocity coupling; displacement and load spectra; analytical plate bending and elastic buckling.

Model development and verification: Conservation-based transient models; coupled 0-D/3-D simulation; parameter-sensitivity studies; comparison with measurements; analytical/FEM cross-checks; mesh and time-step convergence.

Programming and computing: Python, MATLAB, C/C++, Fortran, Mathematica; scientific data analysis and visualisation; HPC simulation workflows using NAISS resources at PDC and NSC.

Engineering software: Simcenter STAR-CCM+, OpenFOAM, COMSOL Multiphysics, ABAQUS, ANSYS, Tecplot, Gnuplot, Inkscape, LaTeX.

Spectral analysis and acoustics: FFT; power and cross-spectral density; coherence and phase; spectral proper orthogonal decomposition (SPOD); acoustic wave and resonance modelling; acoustic–structural coupling; FW–H sound-source analysis.

Research and Professional Experience

Ph.D. Researcher – CFD, Fluid–Structure Interaction and Dynamics Mar 2022 – Present

Department of Engineering Mechanics, FLOW, KTH Royal Institute of Technology, Stockholm

- Develop three-dimensional transient models coupling compressible LES with FEM-based structural mechanics in STAR-CCM+ for upper-airway aeroacoustics, resolving feedback between fluid loads and structural motion through strong two-way FSI and moving overset meshes.
- Compare flow conditions, material stiffness, and geometry variants to determine their effects on structural displacement, force spectra, and vibration; model large deformation using Neo-Hookean hyperelasticity.
- Use COMSOL structural eigenmodes, cross-spectra, coherence, and force–velocity phase analysis to identify modal excitation and explain coupled dynamic response.
- Develop an analytical acoustic resonance model with compliant structural coupling; verify predictions against COMSOL FEM, with frequency differences within 1.03% for the first ten soft-splitter common resonances in the tested geometry (unpublished manuscript).
- Establish numerical reliability through mesh and time-step convergence studies; use pressure/vorticity SPOD to distinguish coherent flow structures from acoustic modes; run simulations on NAISS HPC resources at PDC and NSC.
- Lead analysis, visualisation, and manuscript writing in collaboration with specialists in fluid mechanics, structural dynamics, and acoustics; communicate findings through peer-reviewed publications, including a *Physics of Fluids* Featured Article.

Research Assistant – Industrial Modelling and Compressible CFD Dec 2018 – Mar 2020

Joyson Safety Systems, Aschaffenburg, Germany

- Developed Python-based transient initiator models from mass and energy conservation, combustion, heat transfer, and nozzle-flow relations; generated boundary conditions for coupled CFD simulations.
- Coupled 0-D initiator models to 3-D OpenFOAM simulations to resolve shock-wave propagation and assess inflator performance and structural integrity.
- Compared predicted pressure histories with closed-bomb measurements and evaluated component sensitivities to support design assessment and product development; work published in *Journal of Energy and Power Technology*.

Education

Ph.D. studies – Computational Fluid Dynamics Mar 2022 – Present
KTH Royal Institute of Technology, Stockholm, Sweden
Thesis: *Computational Aero-acoustics of Snoring with Impact on Obstructive Sleep Apnea Diagnosis*
Defence scheduled: 4 December 2026.
Supervision: Mihai Mihaescu, Marco Laudato, and Susann Boij.

M.Sc. in Mechanics (Mechanik) Oct 2016 – Jan 2020
Technical University of Darmstadt, Germany
Final grade: **1.33 (very good)**; thesis: **1.0** (German grading system).
Thesis: *Analysis and modeling of ballistic behaviour of pyrotechnical initiators and their impact in airbag inflators applications.*

B.Eng. in Engineering Mechanics Sep 2011 – Jun 2015
Dalian University of Technology, China
Overall mark: 86/100; thesis mark: 98/100.

Selected Peer-Reviewed Journal Publications

1. **Li, P.**, Laudato, M., and Mihaescu, M. (2026). Computational analysis of palatal non-apneic snoring sound generation using a simplified human upper-airway model. *Physics of Fluids*, **38**(8), 081909. **Featured Article**. doi:10.1063/5.0346307.
2. **Li, P.**, Laudato, M., and Mihaescu, M. (2023). Time-dependent fluid-structure interaction simulations of a simplified human soft palate. *Bioengineering*, **10**(11), 1313. doi:10.3390/bioengineering10111313.
3. **Li, P.**, Chang, C.-Y., and Sadiki, A. (2021). Numerical investigation of pyrotechnical initiators and their impact in airbag inflator applications: Ballistic modeling and functional analysis. *Journal of Energy and Power Technology*, **3**(2), article 026. doi:10.21926/jept.2102026.
4. Wang, B., **Li, P.**, and Li, R. (2016). Symplectic superposition method for new analytic buckling solutions of rectangular thin plates. *International Journal of Mechanical Sciences*, **119**, 432–441. doi:10.1016/j.ijmecsci.2016.11.006.

Selected Unpublished Manuscripts

1. **Li, P.**, Laudato, M., and Mihaescu, M. *Stiffness-dependent modal excitation and force–velocity coupling in fluid-induced soft-palate vibration.* Material-stiffness studies, modal coupling, nonlinear vibration, and acoustic response.
2. **Li, P.**, Laudato, M., and Mihaescu, M. *Effects of tongue-base induced airway constriction on non-apneic palatal snoring.* Geometry-dependent fluid loading, structural response, and pressure/vorticity SPOD.
3. **Li, P.**, Laudato, M., Boij, S., and Mihaescu, M. *Acoustic resonances of a bifurcated duct with a compliant splitter.* Analytical coupled-system modelling, junction impedance, and FEM verification.

Additional Research Experience

Research Assistant – Reduced Combustion Modelling Sep 2020 – Aug 2021
Institute for Technical Thermodynamics, Karlsruhe Institute of Technology, Germany
Applied the reaction–diffusion manifold (REDIM) method to premixed flame–wall interactions; validated reduced kinetic models against detailed kinetics.

Research Project – Low-Mach-Number Flow Modelling Aug – Sep 2018
Institute for Fluid Dynamics, Technical University of Darmstadt, Germany
Used COMSOL FEM to investigate non-isothermal Couette flow and natural convection, comparing low-Mach-number and Boussinesq formulations.

Research Assistant – Analytical and Numerical Solid Mechanics Jul 2013 – Jun 2015
Institute for Solid Mechanics, Dalian University of Technology, China
Derived analytical plate bending and buckling solutions and compared them with FEM using Mathematica, ABAQUS, and MATLAB.

Scientific Communication

Co-authored *Effects of gravity force and tissue stiffness on a simplified three-dimensional human soft palate*, ICSV30, Amsterdam (2024). Conference participation: EFDC2, Dublin (2025); Swedish Mechanics Days, Gothenburg (2024); Swedish Society of Biomechanics, Lund (2024).

Languages

Chinese – Native | English – IELTS 6.5/9.0 | German – TestDaF 17/20
Swedish – A2 (CEFR)